

Health & Safety

Standard Construction machinery and associated maintenance



Reference

Company	Department	Document Type	N°	Version	Langue
BYTP	H&S	REF	2215	C	EN

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A	01-11-2023	O.DANNAU	I.LELIEVRE	L.KNOLL	First issue
B	21-05-2024	A.JANSSENS	I.LELIEVRE O.BRZEZINSKI	L.KNOLL	Removal of 1. fundamentals
C	02-12-2025	F.MARTIN	A.JANSSENS O.BRZEZINSKI	L.KNOLL	Section 4 updated following BYCN Guidelines

The standards for the use of construction machinery presented in this document serve as a practical guide to define the means to be planned and deployed to ensure effective management of the machinery - For sensitive equipment, these elements must be taken into account in addition to the requirements of the "Policy regarding sensitive equipment BYTP and Subsidiaries".

The elements of this document complement the elements presented in the **Standard Pedestrian Machine Interactions Reference (BYTP-H&S-REF-2212)**, dedicated to the organization of machine-pedestrian interfaces. The elements of this document can also serve as support for briefing the various actors involved in an operation involving machines interfacing with pedestrians.

Table of contents

1. Compliance of construction equipment	4.1. Presence of power lines
2. Training and authorization of machine operators	4.2. Presence of buried networks
3. Safety equipment on construction machinery	5. Work on a machine directly on the site
4. Interventions near networks	



1. Compliance of construction equipment

All construction machinery complies with the standards in force in their countries of use.

In others:

- Conformity documents (CE if applicable)
- (VGP) Periodic General Verification
- User manual in the language of the country of use



Certificate of conformity to the European community



Machine user manual in operator's language



Equipment carrying out lifting (hydraulic excavators with hook, handling equipment / forklifts):

- Semi-annual audit
- Machines equipped with a non-return valve

A six-monthly check is mandatory to authorize lifting with a hydraulic excavator.

Lending or borrowing a machine from an external company = establishment of a provision agreement



A closed document box = Uncontrolled documents.

2. Training and authorization of machine operators

The operation of lifting machines or equipment is subject to obtaining a driving authorization issued and signed by the employer or his representative (delegate of authority).

This driving authorization is issued under these 3 conditions:

- A medical fitness to drive test carried out by the occupational physician
- Training on the machine concerned by a person deemed competent
- Specific training (on site) linked to the use of the machine on the site in question

International scope, in the absence of specific training of the CACES® type, training in machine driving with associated test is imperatively carried out

The driving authorization is signed by the operator's company:

- By BYTP's delegated authority for its own staff and temporary workers
- By the delegated authority of the service provider or subcontractor company for their own staff

A driving permit is only issued for a construction site. It is associated with a safety reception on the construction site.

Do not issue an annual driving license.



3. Safety equipment on construction machinery

Minimum safety equipment on construction machinery:

- Seatbelt
- Beacon
- Reversing beep
- Reversing camera(s) or radar required on all vehicles
- Roll bar or cabin
- If necessary, machine-pedestrian detection system



Seatbelt



Beacon



Reversing
beep



Hoop



Camera

CHECK-LIST JOURNALIÈRE

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CONDUCTEUR

NOM : _____

PRÉNOM : _____

PROFESION : _____

SIGNATURE

JE DISPOSE DE MON AGENT ET D'UNE AUTOMATISATION E.C. DE COMMANDE DU CHANTIER ET DE MON APPÉTIT MÉDICALE

A JOUR

LES RISQUES LIÉS À LA COACTIVITÉ ET À LA CIRCULATION SONT PRIS EN COMPTE

LES CLES RESTENT SOLIDAIRES DU CHAUFFEUR

SYSTÈME DE STÉRAGE

☒	Remorquage, Sécurité
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NOTIFIER

☒	Pré et/ou l'équipement capable de déplacement dans les zones Pré de la Zone (ZSL)
☒	Sécher et Nettoyer
☒	Cape

STABILISATION

☒	Stabilisateur
☒	Pneus stabilisés
☒	Pneus stabilisés
☒	Pneus stabilisés
☒	Pneus stabilisés

APERTURES

☒	Coffre
☒	Coffre à clé
☒	Pneus stabilisés
☒	Pneus stabilisés
☒	Pneus stabilisés

DIVERS

☒	Extincteur
☒	Pré à la sécurité

ACCÈS À LA CABINE

☒	Équipement
☒	Niveau d'air

SIGNALISATION

☒	Signalisation
☒	Signalisation
☒	Signalisation
☒	Signalisation
☒	Signalisation

CABINE

☒	Siège
☒	Coffre à clé
☒	Coffre à clé
☒	Coffre à clé
☒	Coffre à clé

REPLACEMENT

☒	Niveau
☒	Niveau
☒	Niveau
☒	Niveau
☒	Niveau



EN CAS DE NON CONFORMITÉ, INFORMER IMMÉDIATEMENT SON CHEF DE CHANTIER

COMMENTAIRES :

Verification of construction equipment before each use by the operator (inspection logbook, use of a mobile application if applicable, etc.).

The operator also checks the presence of the emergency opening ring or the presence of a window breaker hammer.



The use of mobile phones or radio/music is prohibited outside the parking areas provided for this purpose. On site, any use of telephone or radio type equipment is prohibited while driving.

In the event of increased coactivity between machines or between machines and pedestrians, additional presence detectors or cameras may be necessary.

Refer to reference document **BYTP-H&S-REF-2212 (Pedestrian Machine Interactions Reference)**.

If the machine is stopped, the key remains attached to the driver.



4. Interventions near networks

Intervention activities near aerial or buried networks are risky activities.

It is necessary to ensure that all networks are known and marked before any intervention near them.

It is necessary to ensure that the concessionaires of these networks are known, informed of upcoming activities and able to intervene in the event of damage to their networks.

Machine and foot operators working near networks:

- Are trained in the identification of these networks
- Are informed of the networks present
- Received on-the-job training before starting their activities
- Are holders of an authorization to intervene near networks

It is necessary to keep up-to-date a utility network plan that specifies the location of all overhead and buried powerlines within or in the immediate vicinity of the project, taking into consideration:

- Detailed plans provided by public services and energy provider,
- Additional checks carried out with the use of the underground cable locator,
- The location of lines specifically installed for the project's power supply.

In the case proper plans cannot be provided for underground services, the project is equipped with underground cable locator and checks are carried out within and in the immediate vicinity of the project

Authorization for intervention near networks:

The authorization for intervention by the delegate of authority of the site can only be issued to a collaborator whose competence is confirmed by one of the following elements:

- a valid training certificate integrating intervention near networks (a CACES® for the example of France)
- a title, diploma or professional qualification certificate for public works professions less than 5 years old
- a certificate of skills obtained from an examination center after passing the exam valid for 5 years

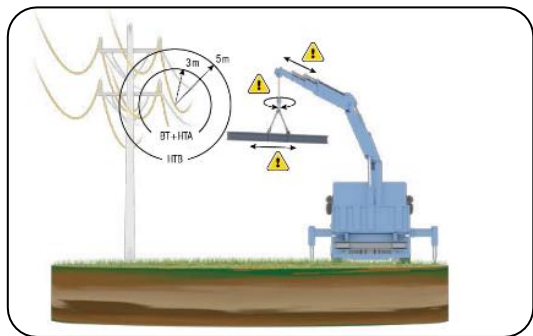
Apply any more restrictive local regulations

Authorization for intervention near networks is necessary for:

- Designers (Technical direction in case of design work)
- Supervisors (work supervision, master's degree, assistant and aspiring leader)
- Operators (companions, machine and crane drivers, team leaders, layers, pipeliners)

4.1. Presence of power lines

In the event of the presence of electrical lines near moving machinery or used for lifting loads, it is essential to avoid any contact or ignition with the active parts of non-insulated electrical installations or any damage to the surrounding electrical installations.



The minimum safety distances to be respected are as follows:

- If $T < 50,000 \text{ V}$ = Safety distance > 3 meters
- If $T > 50,000 \text{ V}$ = Safety distance > 5 meters

The possible swinging as well as the size of the load are dimensional.

Apply any local regulations
more restrictive



Overhead lines are identified on the Site Installation Plan (PIC) and physically identified on site with safety templates when necessary.

4.2. Presence of buried networks

Taking into account buried networks is a key element in the use of machines. It is right to :

- Identify their locations before starting work (marking-staking)
- Take their locations into account when choosing the location of lifting means
- Take their locations into account when carrying out earthworks and trenches
- Know what to do in the event of a network emergency
- Be guided in the event of mechanical earthworks less than 1.50m from a network



Electricité BT, HTA ou HTB et éclairage		Rouge
Gaz combustible (transport/distribution), hydrocarbures		Jaune
Produits chimiques		Orange
Eau potable		Bleu
Assainissement et Pluvial		Marron
Chauffage et Climatisation		Violet
Télécommunications		Vert
Feux tricolores et Signalisation routière		Blanc
Zone d'emprise multi-réseaux		Rose

The incorrect positioning of lifting equipment or an earthmoving activity can lead to the rupture of a network by puncture or shock leading to the release of pressurized gas, high voltage electricity or other sources of dangerous energy.

5. Work on a machine directly on site – 1/3

Before leaving for intervention:

- Find out about the nature of the intervention (type of breakdown), and try to collect as much information as possible
- Consult the history of the machine (logbook, CMMS) and the manuals available
- Consult the procedures manual for routine maintenance operations
- Plan your intervention first with the chief engineer then with the site manager

Prepare your intervention:

- Make sure not to use damaged or non-compliant equipment
- Check your PPE before leaving for intervention (at least helmet with chin strap and safety cap, gloves, long-sleeved clothing, safety shoes, protective glasses and hearing protection)
- Consult the SDS, the Product Data Sheet and the job description for any use of a dangerous chemical product
- Store your van (unload unnecessary parts or tools, waste) in order to anticipate any overload
- Store worn and new parts separately
- Ensure the presence of all necessary equipment (environment kit/empty bin/first aid kit)



First aid kit



Environmental kit



A torque wrench is checked visually before each use and every 2 years by an external organization.



Defective tool

5. Work on a machine directly on site – 2/3

When arriving on site:

- Compulsorily follow the welcome/induction visit during the first intervention on the site
- Communicate with the person responsible for the work and mechanics of the machine location depending on the type of intervention (analyze the risks, traffic conditions, safety conditions)
- Collect the logbook and the machine manual (supplemented when possible by computer-assisted maintenance management - CMMS) in order to be guided during repairs
- Provide sufficient space, clean, stable and not intrusive

Note: A maintenance zone is necessarily indicated on the PIC

Secure the intervention area:

- If the machine can be moved, ask the driver to position it in a secure area
- If the machine cannot be moved, mark the intervention area (cone, tape, etc.)

Note: Respect the conventional signs to guide the machine and respect the rules for machine/pedestrian interactions

Equipment logging:

- Lower the equipment before any intervention (unless otherwise provided by the manufacturer, or impossible)
- OBLIGATION to put one or more blocks on the machine, turn the battery switch to “off”, secure it with a padlock and remove the keys from the ignition
- Find out if the driver recommends special instructions (driving and maintenance manual)
- Affix your lockout badge prominently on the machine during the intervention
- Walk around the machine to determine other potential risks and problems (leaks, abnormal heating, risk of falling, crushing)



Put an “Out of service” label in the event of stopping the intervention, record it if necessary, until the machine is put back into operation

5. Work on a machine directly on site – 3/3

Working environment :

- Intervene in a clean manner (avoid any pollution)
- Consider using the locking pins provided for this purpose for certain operations (e.g. for tippers or truck or loader joints)
- Ensure that no waste remains after an intervention (absorbents, plastic, traces of oil, etc.)
- Use the corresponding tools and PPE (check their good condition)
- Read the instructions for use and inquire about the risks with your mechanical manager before using specific tools.
- Use the tools in accordance with their primary use



Gestures and postures :

- Use the correct working positions
- Access and descend in front of the machines using the 3 point of support rule (non-slip surfaces required on handles and steps)
- Prohibition on jumping off machines to get off them
- Prohibition to work on the edges of a machine in the absence of guardrails or collective protection provided
- If necessary, dismantle certain elements to access “restrictive” parts of the machines
- During long and/or repetitive interventions, organize your work plan methodically (e.g.: tightening tiles, changing teeth, long dismantling operation)

